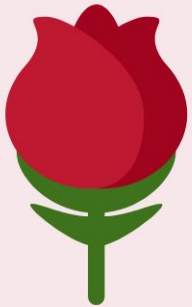
Jargon, Acronyms, and Astrobites

TIMESTEP Friday Workshop

ROSE



Something positive from this week.

THORN



Something difficult from this week.

BUD



Something you're looking forward to next week.

What is jargon?

- Specialized and/or technical language used by a particular field or group that outsiders typically don't understand

Acronyms

- From NASA to JWST to GRMHD, astronomers love acronyms

What are some examples you've encountered so far?

- Add them to your white boards!

Why do astronomers use so much jargon?

- Sometimes it’s simpler and faster than having to explain yourself
- Astronomers spend a lot of time talking to other subject matter experts
- Jargon can make you sound like you’re at a high level (sort of iykyk situation)

What are some problems with using jargon and acronyms?

- It can limit understanding! If people don’t even know the words you’re using they won’t understand your science
  - This is bad for collaboration!!!
- It can make environments seem less welcoming to undergrads or other people who are new to the field
- Jargon puts up entrance barriers

Abstract Example - What jargon do you see here?

A 33 M⊙ black hole (BH) was recently discovered in an 11.6-year binary only 590 pc from the Sun. The system, Gaia BH3, contains a 0.8 M⊙ low-metallicity giant ([M/H] =−2.2) that is a member of the ED-2 stellar stream. This paper investigates whether the system could have formed via isolated binary evolution. I construct evolutionary models for metal-poor massive stars with initial masses ranging from 35−55 M⊙, which reach maximum radii of 1150−1800 R⊙ as red supergiants. I then explore what combinations of initial orbit, mass loss, and natal kick can produce the period and eccentricity of Gaia BH3. Initial orbits wide enough to accommodate the BH progenitor as a red supergiant can match the observed period and eccentricity, but only if the BH formed with a significant natal kick ($v_{kick} \gtrsim 20$ km s^{-1}). These models are disfavored because such a kick would have ejected the binary from the ED-2 progenitor cluster. I conclude that Gaia BH3 likely formed through dynamical interactions, unless the BH progenitor did not expand to become a red supergiant. Only about 1 in 10,000 stars in the solar neighborhood have metallicities as low as Gaia BH3. This suggests that BH companions are dramatically over-represented at low-metallicity, though caveats related to small number statistics apply. The fact that the luminous star in Gaia BH3 has been a giant – greatly boosting its detectability – only for~1% of the time since the system's formation implies that additional massive BHs remain to be discovered with only moderately fainter companions. Both isolated and dynamically-formed BH binaries with orbits similar to Gaia BH3 are likely to be discovered in Gaia DR4.

Rewrite your group's section in more accessible language

A 33 M⊙ black hole (BH) was recently discovered in an 11.6-year binary only 590 pc from the Sun. The system, Gaia BH3, contains a 0.8 M⊙ low-metallicity giant ([M/H] =−2.2) that is a member of the ED-2 stellar stream. This paper investigates whether the system could have formed via isolated binary evolution. I construct evolutionary models for metal-poor massive stars with initial masses ranging from 35−55 M⊙, which reach maximum radii of 1150−1800 R⊙ as red supergiants. I then explore what combinations of initial orbit, mass loss, and natal kick can produce the period and eccentricity of Gaia BH3. Initial orbits wide enough to accommodate the BH progenitor as a red supergiant can match the observed period and eccentricity, but only if the BH formed with a significant natal kick ($v_{kick} \gtrsim 20$ km s−1). These models are disfavored because such a kick would have ejected the binary from the ED-2 progenitor cluster. I conclude that Gaia BH3 likely formed through dynamical interactions, unless the BH progenitor did not expand to become a red supergiant. Only about 1 in 10,000 stars in the solar neighborhood have metallicities as low as Gaia BH3. This suggests that BH companions are dramatically over-represented at low-metallicity, though caveats related to small number statistics apply. The fact that the luminous star in Gaia BH3 has been a giant – greatly boosting its detectability – only for~1% of the time since the system's formation implies that additional massive BHs remain to be discovered with only moderately fainter companions. Both isolated and dynamically-formed BH binaries with orbits similar to Gaia BH3 are likely to be discovered in Gaia DR4.

Key Takeaways

- By spending some time to break down the abstract we can now understand it! Yay!
  - You can apply this to the papers you read in your research project
- Eliminating the jargon made the abstract less concise
  - When writing for experts (like writing a scientific paper) it's okay to use some jargon and acronyms, but make sure you define them!
- When talking to non-experts about your research project try to limit the amount of jargon and acronyms you use. Lower the entrance barrier for those around you

Introducing: TIMESTEP Astro Glossary

- Pull up the document I sent in Slack
- Work with your table groups to add some words from your whiteboards



TIMESTEP Astro Glossary!

Add the definitions of any astronomy words or acronyms that you don't know below

✨ **Galaxies**

✨ **Stars**

✨ **Instrumentation/Telescopes**

✨ **Cosmology**

✨ **Compact Objects**

Where do we go to find the meanings of jargon and acronyms?

- Google it!
  - The NASA and ESA (European Space Agency) websites have lots of articles explaining research topics and ongoing missions
  - Wikipedia
- Ask your research advisor, grad mentor, or another student in your research group
- Astrobites!

So yes, reading papers is challenging… (especially when first starting out!)

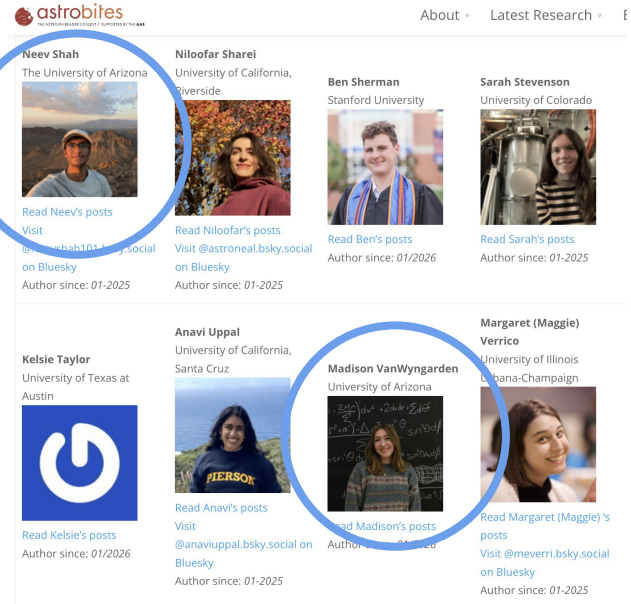


astrobites

THE ASTRO-PH READER'S DIGEST | SUPPORTED BY THE **AAS**

- Daily summaries of astronomy papers
  - like a blog post!
- Primarily aimed at undergrads (like you!)
- Reduce the entry barrier to reading papers

who writes astrobites?



+ many more

- graduate students from across the world (you might recognize a couple of them)
- one interesting paper per day where you learn something new
  - what's the broad research question? why is it interesting?
  - what techniques were used?
  - explain the jargon used in the paper!

let's look at an example

we saw this earlier - even the abstract can be hard to read if you're not already an expert

A 33 M⊙ black hole (BH) was recently discovered in an 11.6-year binary only 590 pc from the Sun. The system, Gaia BH3, contains a 0.8 M⊙ low-metallicity giant ([M/H] =−2.2) that is a member of the ED-2 stellar stream. This paper investigates whether the system could have formed via isolated binary evolution. I construct evolutionary models for metal-poor massive stars with initial masses ranging from 35−55 M⊙, which reach maximum radii of 1150−1800 R⊙ as red supergiants. I then explore what combinations of initial orbit, mass loss, and natal kick can produce the period and eccentricity of Gaia BH3. Initial orbits wide enough to accommodate the BH progenitor as a red supergiant can match the observed period and eccentricity, but only if the BH formed with a significant natal kick ($v_{kick} \gtrsim 20$ km s−1). These models are disfavored because such a kick would have ejected the binary from the ED-2 progenitor cluster. I conclude that Gaia BH3 likely formed through dynamical interactions, unless the BH progenitor did not expand to become a red supergiant. Only about 1 in 10,000 stars in the solar neighborhood have metallicities as low as Gaia BH3. This suggests that BH companions are dramatically over-represented at low-metallicity, though caveats related to small number statistics apply. The fact that the luminous star in Gaia BH3 has been a giant – greatly boosting its detectability – only for~1% of the time since the system's formation implies that additional massive BHs remain to be discovered with only moderately fainter companions. Both isolated and dynamically-formed BH binaries with orbits similar to Gaia BH3 are likely to be discovered in Gaia DR4.

there is an astrobite on the same paper!

astrobites
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Too big to fit? Recipes on how (not) to create the black hole binary that you find in your backyard

by Neev Shah | Feb 17, 2025 | Daily Paper Summaries | 1 comment

Title: On the formation of a 33 solar-mass black hole in a low-metallicity binary

Authors: Kareem El-Badry

First Author's Institution: Department of Astronomy, California Institute of Technology, 1200 E. California Blvd., Pasadena, CA 91125, USA

Status: Published open access in The Open Journal of Astrophysics.

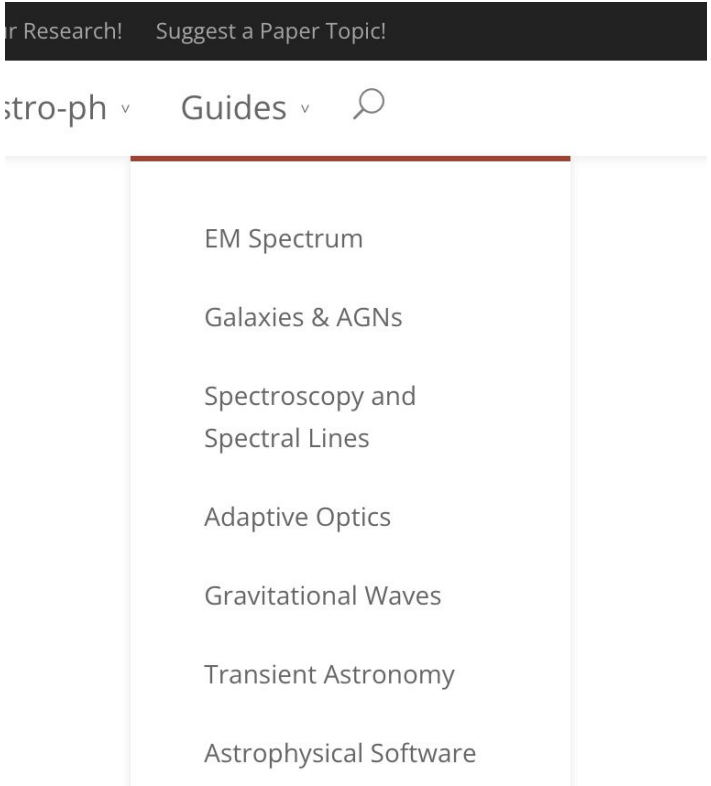
Black holes are ubiquitous throughout astronomy, from the supermassive giants lying in the center of most galaxies, to their stellar mass counterparts formed when massive stars die. For decades, we've found the latter in X-ray binaries, and more recently through gravitational wave signals. However, the last two years have seen a new black hole discovery machine: everyone's favourite space astrometry satellite, Gaia. Since 2013, Gaia has been observing the motions of over a billion stars in our galaxy! A significant fraction of them live in binary systems, which causes tiny wobbles in their motion that Gaia can measure. Their shape, size, and duration can reveal key information about the binary, such as the orbital period and component masses.

Some of the papers you are reading may* already have an astrobite!

*there are on average ~100 new papers *every day* on the arXiv astro-ph section

there is a lot of other useful resources on astrobites

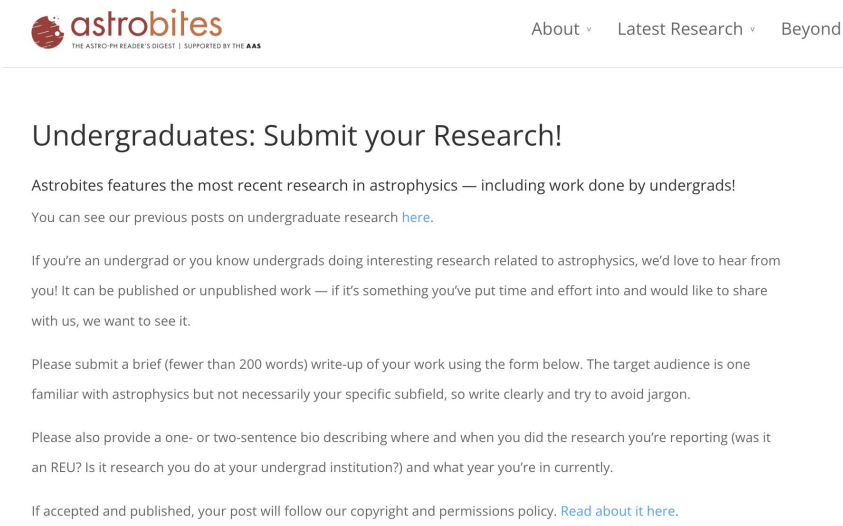
- do you want to learn about all the different wavelengths in the electromagnetic spectrum (or gravitational waves)?
- need to learn how to use LaTeX? or DS9 to view your astronomical images?
- how to apply for an REU? or grad school?



there may already be an astrobites guide to help you get started

You can submit your research too!

- if you want to share your research to a wider audience, astrobites is a great place to showcase all the amazing work that all of you will do over the next few months!
- does not need to be published!
- (scientific) writing, and communicating your work is an important part of any career, including astronomy
  - summarizing your research can also help improve your own understanding, and on how to avoid jargon



activity to try out in your free time

- check if any of the papers that your advisor has given you already has an astrobite

- think about the broader field that you are working in
  - try explaining (~in a minute) to your neighbour what the term means, and why is it interesting?
  - look up the terms on astrobites to see if you can find any interesting bites on the topic!
  - E.g., - “molecular clouds”, “protoplanetary disks”, “binary stars”